

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

With Magnetic Base Drill

- Crankshaft adapter and gear remover kit, Part Number 5572864
- Magnetic base drill, Part Number 5572871 (60 Hz) or 5572872 (50 Hz)
- Cutter bit insert, Part Number 5572875
- Gear remover, Part Number ST-647

Without Magnetic Base Drill

- Plastic drill guide, Part Number 4918547
- Stop collar, Part Number 4918541
- Large drill guide adapter, Part Number 4918276
- Alignment pin, Part Number 4918577
- Countersink cutter, Part Number 4918578
- Drill bushing, Part Number 4918534
- Locating gauge, Part Number 4918314
- Small splitting wedge, Part Number 4918545
- Locating gauge, Part Number 4918314

Additional Service Items

- Crocus cloth to remove any scratches or nicks

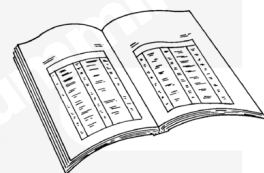
Preparatory Steps

On-Engine Repair

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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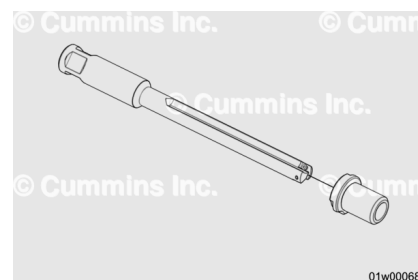
ck800wa

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Remove the lower gear cover. Refer to Procedure 001-080 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-080-tr.html>)
- Remove the crankshaft adapter. Refer to Procedure 001-017 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-017-tr.html>)
- Remove the tone wheel. Refer to Procedure 001-069 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-069.html>)
- Remove the alignment dowel pin.
- Remove the lubrication oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>)
- Remove the lubrication oil pump. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html>)
- Remove the rocker lever cover and gasket. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>)
- Disconnect and remove the engine brake wiring harness. Refer to Procedure 020-015 in Section 20. (</qs3/pubsys2/xml/en/procedures/10/10-020-015.html>)
- Remove the rocker lever shaft. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/10/10-003-009.html>)
- Remove the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html>) Remove the valve camshaft gear, injector camshaft gear, adjustable idler gear, and the lower idler gear. Refer to Procedure 001-088 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html>)

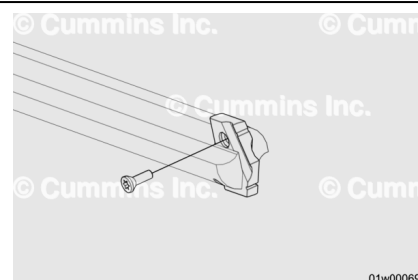
Remove

With magnetic base drill:

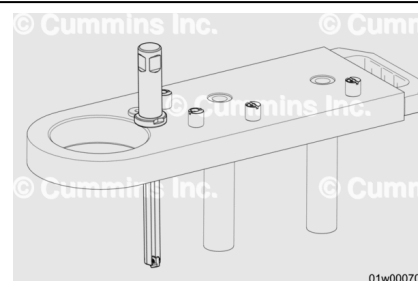
Slide the removable drill guide, Part Number 4918539, from guide plate kit, Part Number 5572686, onto the cutter bit holder, Part Number 5572874.



Install the 5/8" cutter bit insert, Part Number 5572875, into the cutter bit holder, Part Number 5572874 and torque to 1.75 N•m [15.5 in-lb].

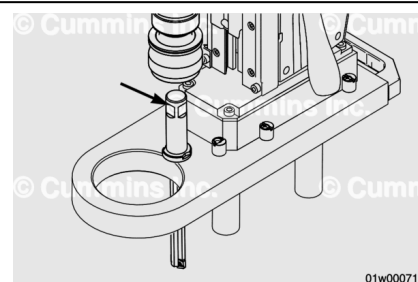


Install the drill bit holder assembly into the crankshaft gear remover plate, Part Number 5572686, and lock the guide into place.



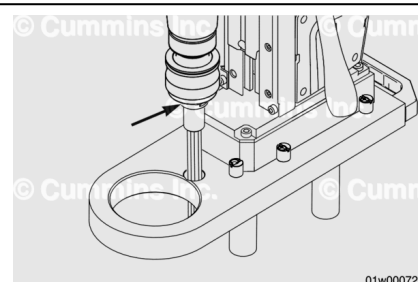
Place the magnetic base drill onto the crankshaft gear remover plate, Part Number 5572686, close to the bottom and side alignment stop blocks.

Make sure the drill base is close to the alignment stop blocks. The alignment stop blocks will be adjusted later to maintain the drill in the set position.



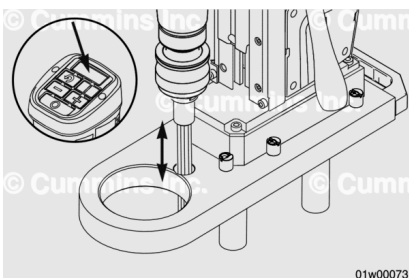
Install cutter bit holder into the magnetic base drill tool bit holder.

Make sure magnetic base drill tool bit holder is locked into position. The slide lock will be all the way up. You will hear or feel a click when it is locked into place.



Turn the magnet on.

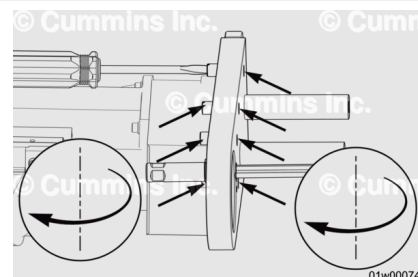
Verify the cutter bit holder moves freely in the drill guide.



Turn the magnetic base drill and crankshaft gear remover plate assembly onto its side, exposing the alignment stop block fasteners.

Loosen the fasteners.

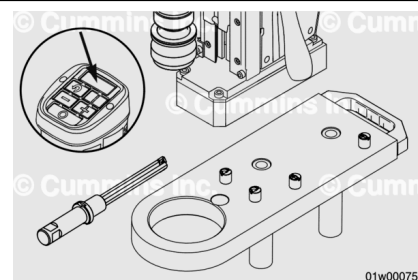
Use a flat tip screwdriver to adjust the alignment stop blocks snug to the drill base, then tighten the fasteners.



Remove the cutter bit holder from the magnetic base drill bit holder.

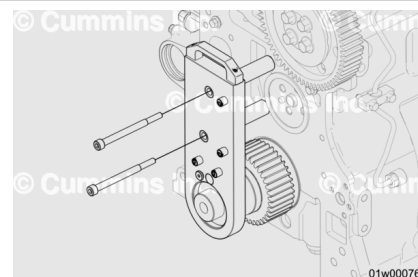
Turn off the magnet and remove the magnetic base drill from the crankshaft gear remover plate.

Remove the cutter bit assembly from the crankshaft gear remover plate.

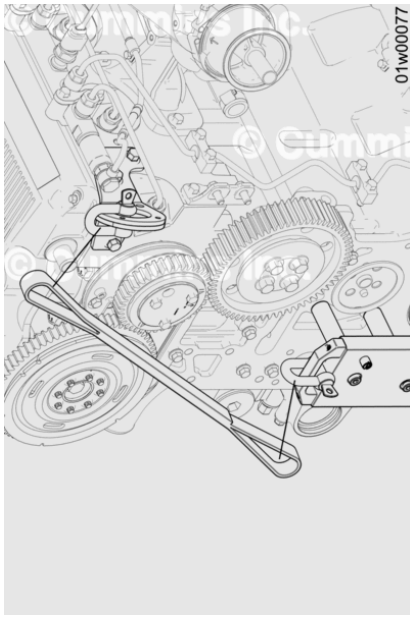


Install the crankshaft gear remover plate, Part Number 5572686, onto the crankshaft using two socket head capscrews, Part Number 5258330.

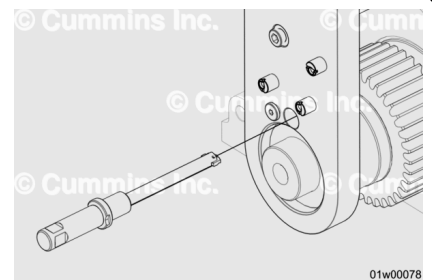
Torque Value: 27 n•m [240 in-lb]



Install the safety strap, Part Number 5299864, from the gear remover plate to the front lifting bracket with two clevis, Part Number 5299865.

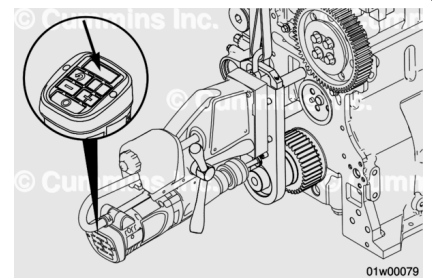


Install the previously assembled cutter bit assembly into the drilling guide plate and lock into place.

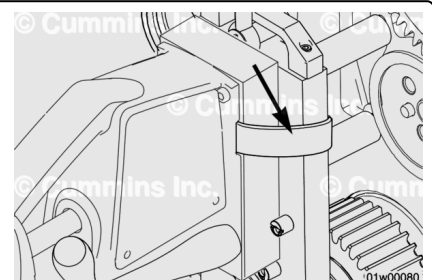


Place the drill onto the guide plate against the bottom and side alignment blocks.

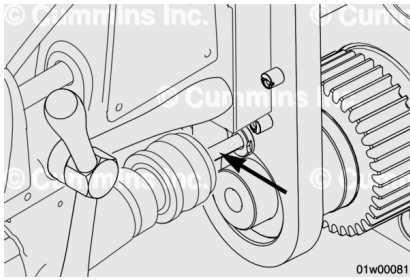
Turn the magnet on.



Install a safety strap to secure the drill to the plate.

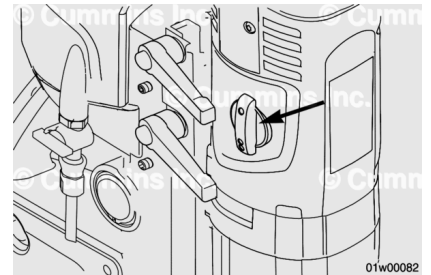


Install the drill tool holder into the drill assembly and lock into place. You will hear or feel a click when the drill tool holder is locked into place correctly.

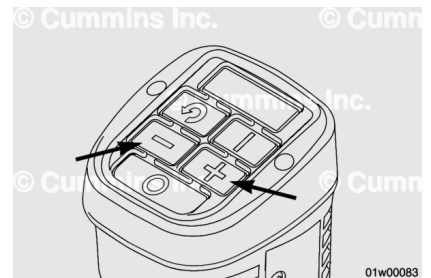


Verify the cutter moves freely in the guide.

Switch the gear setting of the drill to slow, position one.

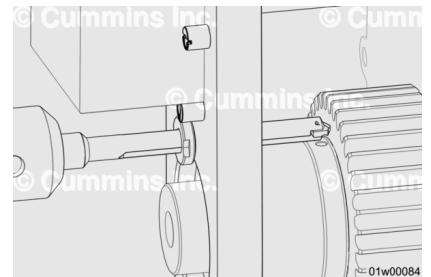


Turn the drill on and adjust the speed to approximately 260 rpm by pressing the minus button repeatedly until it reaches the slowest speed, then press the plus button once.

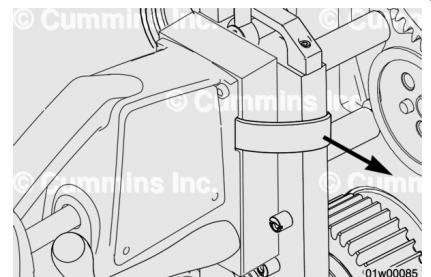


Feed the cutter into the drill guide until the cutter is touching the crankshaft gear.

Apply light pressure to the drill until the cutter gets through the hardened material, then use moderate pressure until the drill travel bottoms out or the gear cracks.



Back the cutter out of the gear and turn the drill off. Remove the drill bit holder from the drill. Remove the safety strap holding the drill to the gear remover plate.

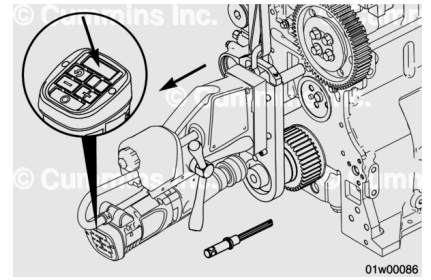


Hold firmly on to the drill and turn the magnet off.

Remove the drill from the gear remover plate.

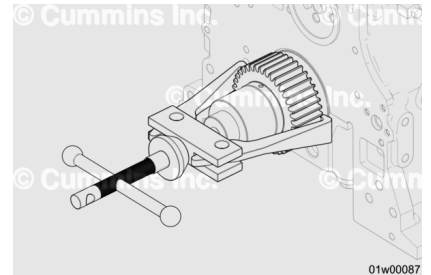
Remove the cutter bit holder assembly from drill guide plate.

Remove the crankshaft gear remover plate and the safety strap from the engine.



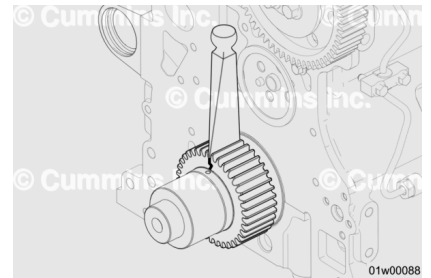
Install a large two or three jaw puller onto the crankshaft gear.

Remove the gear from the crankshaft.



If the gear does not move easily with the puller, check to see if the gear has cracked

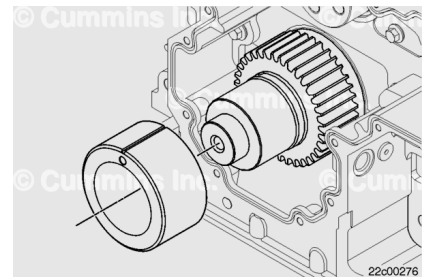
If the gear is not cracked, it may be necessary to use a wedge at the outside between the gear teeth to split the gear.



Without magnetic base drill:

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

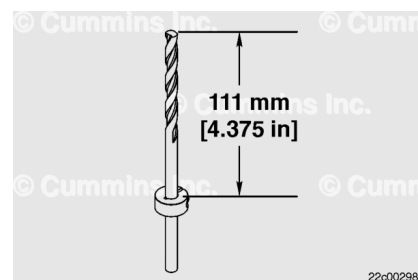


⚠ CAUTION ⚠

Protect the engine internals from debris generated during the drilling operation. Mask the engine internals using tape, plugs, and or shields to prevent debris from entering the engine internals. Failure to do this will result in significant engine damage.

Install the plastic drill guide, Part Number 4918547, with the "FRONT" facing the operator. Align the slot with the crankshaft gear tooth root.

Install the stop collar, Part Number 4918541, or equivalent, onto a 6.35 mm [0.250 in] drill bit at a depth of 111 mm [4.375 in].

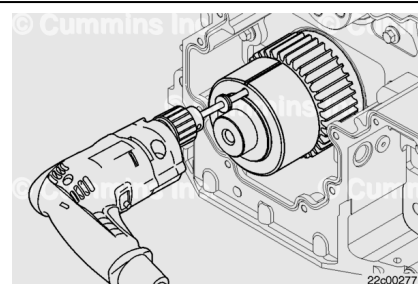


Tool life will be extended if the drill bit does **not** break through the back of the crankshaft gear.

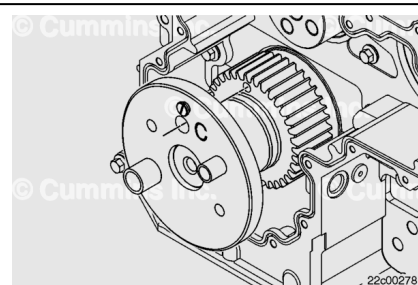
The use of a generous amount of cutting fluid will increase tool life and ease the drilling operation.

Drill a 6.35 mm [0.250 in] pilot hole to the preset depth.

Clean out the debris. Use low pressure air and a shop towel to cover the hole as necessary.

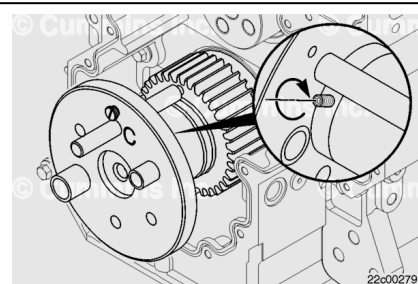


Install the large drill guide adapter, Part Number 4918276, onto the crankshaft nose with the hole labeled "C" aligned with the pilot hole.



Use the alignment pin, Part Number 4918577, to align the guide with the hole.

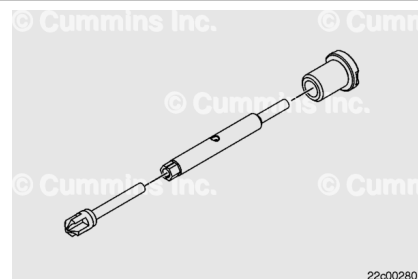
Tighten the 7.94 mm [0.295 in] set screw to hold the guide to the crankshaft.



Install the countersink cutter, Part Number 4918578, or equivalent, into a 102 mm [4.02 in] extension, Part Number 4918544 or equivalent.

Slide the 11.5 mm [0.453 in] drill bushing, Part Number 4918534, or equivalent, onto the shaft.

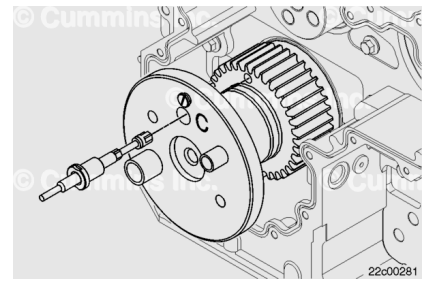
Install the assembly and lock the 11.5 mm [0.453 in] drill bushing into the hole labeled "C".



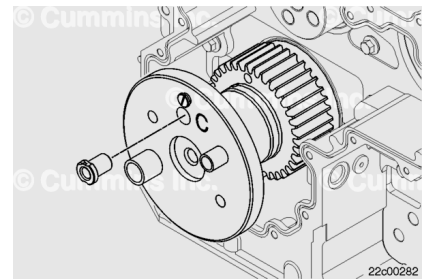
Countersink the pilot hole until the outside diameter of the countersink is approximately 1.6 to 3.2 mm [0.062 to 0.126 in] deep.

Clean out the debris. Use low-pressure air and a shop towel to cover the hole as necessary.

Remove the countersink and the drill bushing.



Insert the drill bushing, Part Number 4918539, or equivalent, into the hole labeled "C" and lock the 12.7 mm [0.50 in] drill bushing into the drill guide.

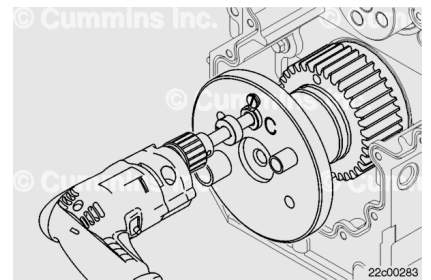


The use of a generous amount of cutting fluid will increase tool life and ease the drilling operation.

Drill a 12.7 mm [0.50 in] hole to depth of the previously 6.35 mm [0.250 in] drilled hole.

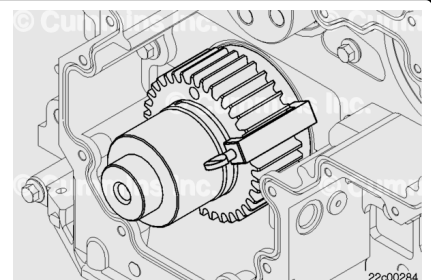
Clean out the debris. Use low pressure air and a shop towel to cover the hole as necessary.

Remove the tone wheel alignment dowel from the crankshaft.

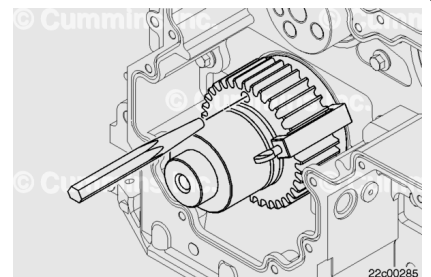


Install the locating gauge, Part Number 4918314, or other suitable spacer, onto the crankshaft gear.

A spacer **must** be installed or the crankshaft will be damaged during the crankshaft gear removal.

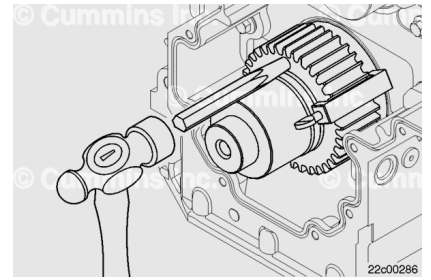


Insert the small splitting wedge, Part Number 4918545, with the flatted area facing toward the crankshaft and into the drilled hole.



Strike the small splitting wedge with a suitable hammer until the crankshaft gear splits.

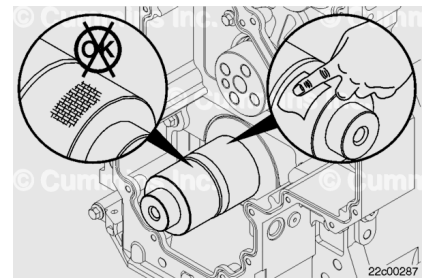
Remove the crankshaft gear by hand or use a suitable puller, if necessary.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a lint-free cloth and solvent to clean any Loctite™ 620 residue from the nose of the crankshaft.

Use a crocus cloth to remove any scratches or nicks.

Note : If scratches or nicks can be felt with a fingernail after the crankshaft has been polished with a crocus cloth, the crankshaft must be replaced.

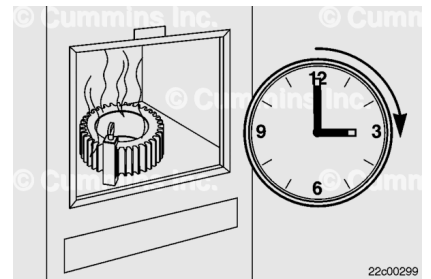
Inspect the crankshaft for damage.

Inspect the machined surface for scratches or nicks.

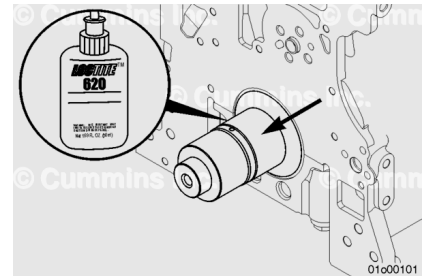
Install

Install the locating gauge, Part Number 4918314, onto the side of crankshaft gear.

Heat the crankshaft gear for 3 hours at 177°C [350°F] before attempting the installation.



Apply two drops of Loctite™ 620 at the 10 o'clock and 2 o'clock locations on the crankshaft where the crankshaft gear will be installed.



⚠ WARNING ⚠

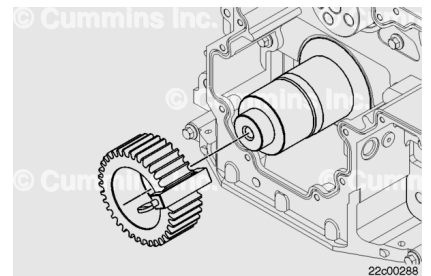
To reduce the possibility of personal injury, wear protective gloves when handling parts that have been heated.

Note : The crankshaft gear must be installed within 15 seconds after being removed from the oven.

Install the crankshaft gear until the locating gauges make contact with the front crankshaft journal.

Hold the crankshaft gear in position for approximately 60 seconds to allow the shrink fit to properly hold the components.

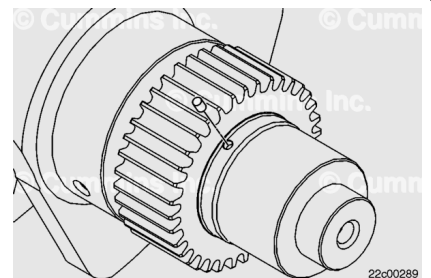
Remove the locating gauge from the crankshaft gear.



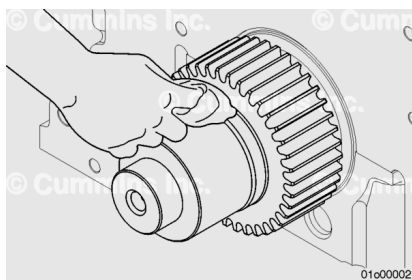
Use a brass hammer to tap the alignment dowel pin in the crankshaft hole until the alignment dowel pin bottoms out.

Check the gap between the crankshaft gear and the crankshaft first journal by inserting the locating gauge, Part Number 4918314.

If the locating gauge is loose or can **not** be inserted, replace the crankshaft gear.



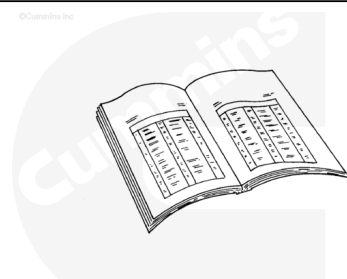
Remove any visible Loctite™ from the crankshaft with a damp towel.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the valve camshaft gear, injector camshaft gear, adjustable idler gear, and the lower idler gear. Refer to Procedure 001-088 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html>)
- Install the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html>)
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- Install the lubrication oil pump. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html>)
- Install the lubrication oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>)

007-025-tr.html)

- Install the tone wheel. Refer to Procedure 001-069 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-069.html)
- Install the crankshaft adapter. Refer to Procedure 001-017 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-017-tr.html)
- Install the lower gear cover. Refer to Procedure 001-080 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-080-tr.html)
- Connect the batteries. Refer to the OEM service manual.

Last Modified: 01-Feb-2022
